

Non-equilibrium critical phenomena: Directed Percolation (and more)

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Spreading processes as models of non-equilibrium critical phenomena¹

Spreading phenomena are characterized by two competing processes: (1) offspring/infection and (2) death/recovery

Depending on their relative rates:

- ① stationary active state (balance between competing processes)
- ② absorbing state (extinction)

This is a genuine non-equilibrium process (no detailed balance):

- Finite systems: eventually reach the absorbing state.
- Infinite systems: (continuous) non-equilibrium transition between active and inactive “phases”.

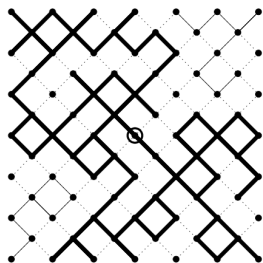
Exploiting scaling invariance (and RG), we can identify critical exponents.

¹Content and pictures are mostly taken from the review by H. Hinrichsen, *Nonequilibrium Critical Phenomena and Phase Transitions into Absorbing States*, Adv. in Phys. (2000), see <https://arxiv.org/abs/cond-mat/0001070>.

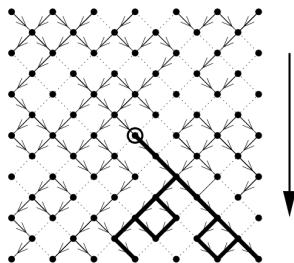
From isotropic to directed percolation

Ordinary (bond) percolation is isotropic (through open bond with prob p).

Directed percolation (DP) is an anisotropic (bond) percolation.



isotropic bond percolation



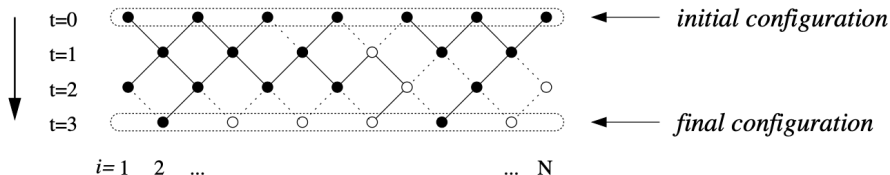
directed bond percolation

Phase transition: probability that a random site belongs to the infinite cluster is $P(p) = 0$ for $p < p_C^{DP}$ and $P(p) > 0$ for $p > p_C^{DP}$

Connected clusters subsets of those of isotropic percolation ($p_C^{DP} > p_c$).

Directed percolation as a spreading process

DP can be interpreted as a spreading process in $d + 1$ dimensions.

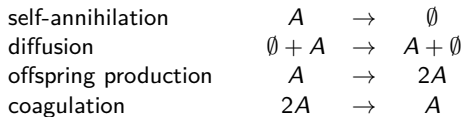


Diagonal lattice, parallel update, $z_i^\pm \sim U[0, 1]$, config. $s(t) = \{s_i(t)\}_{i=1}^N$

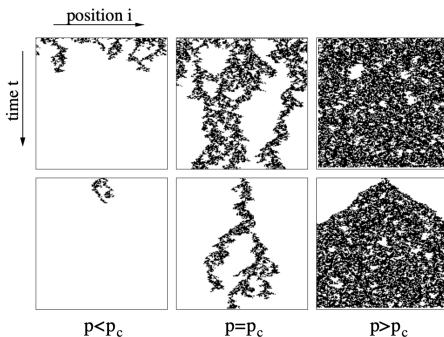
$$s_i(t+1) = \begin{cases} 1 & \text{if } s_{i-1}(t) = 1 \text{ and } z_i^- < p \\ 1 & \text{if } s_{i+1}(t) = 1 \text{ and } z_i^+ < p \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Mapping onto reaction-diffusion processes

Active sites are mapped on particles A , inactive sites on vacancies $\emptyset$,



REMARK: Absorbing states in RD processes if no $\emptyset \rightarrow A$.



Scaling theory: phenomenological derivation

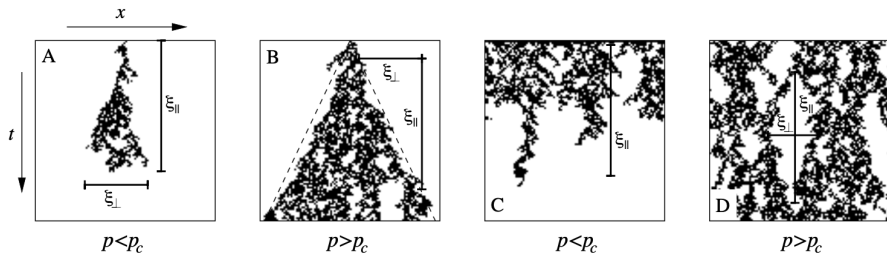
Density of active sites $\rho(t) = \left\langle \frac{1}{N} \sum_i s_i(t) \right\rangle$ (order parameter)

Survival probability $P(t)$ that a randomly chosen site is active (belongs to infinite cluster).

At stationarity: $\rho^{\text{st}} \sim (p - p_c)^\beta$ and $P(\infty) \sim (p - p_c)^{\beta'}$ (for DP $\beta = \beta'$).

Correlation length: spatial $\xi_\perp \sim |p - p_c|^{-\nu_\perp}$, temporal $\xi_\parallel \sim |p - p_c|^{-\nu_\parallel}$.

The *dynamic exponent* is $z = \nu_\parallel / \nu_\perp$.



Scaling theory: phenomenological derivation

Assuming scaling laws ($\Delta = p - p_c$)

$$\rho(t) \sim t^{-\alpha} f\left(\Delta t^{1/\nu_{\parallel}}\right), \quad P(t) \sim t^{-\delta} g\left(\Delta t^{1/\nu_{\parallel}}\right) \quad (2)$$

with

$$f(x) \sim \begin{cases} \text{const} & x \rightarrow 0 \\ x^{\alpha\nu_{\parallel}} & x \rightarrow \infty \end{cases} \quad g(x) \sim \begin{cases} \text{const} & x \rightarrow 0 \\ x^{\delta\nu_{\parallel}} & x \rightarrow \infty \end{cases} \quad (3)$$

In the stationary regime,

$$\rho^{\text{st}} = \rho(\infty) \sim \Delta^{\alpha\nu_{\parallel}}, \quad P = P(\infty) \sim \Delta^{\delta\nu_{\parallel}} \quad (4)$$

but also $\rho^{\text{st}} \sim \Delta^{\beta}$ and $P \sim \Delta^{\beta'}$, then

$$\alpha = \beta/\nu_{\parallel}, \quad \delta = \beta'/\nu_{\parallel}. \quad (5)$$

Scaling theory: phenomenological derivation

Pair correlation function $C(\vec{x}, t)$: probability that site $(\vec{x}, t)$ is connected by a directed path of open bonds to $(\vec{0}, 0)$. It is observed that

$$C(\vec{x}, t) \sim t^{\theta-d/z} F\left(x/t^{1/z}, \Delta t^{1/\nu_{\parallel}}\right) \quad (6)$$

At stationarity,

$$C(0, \infty) = \Delta^{\nu_{\parallel}(d/z-\theta)}, \quad (7)$$

but also $C(0, \infty) \sim \rho^{\text{st}} P(\infty) \sim \Delta^{\beta+\beta'}$.

It gives the *generalized hyperscaling relation* (only for $d \leq d_{\text{uc}}$)

$$\theta - \frac{d}{z} = -\frac{\beta + \beta'}{\nu_{\parallel}}. \quad (8)$$

REMARK: exponent θ says how the average number of active sites grows with t at initial times.

Universality class: the DP conjecture

Janssen (1981) and Grassberger (1982) proposed the *DP conjecture*:
a model should belong to DP universality class if

- 1 continuous phase transition from an active phase into a unique absorbing state;
- 2 transition is described by a positive one-component order parameter;
- 3 dynamic rules involve only short-range processes;
- 4 no disorder, no additional symmetries and conserved quantities.

Examples: contact process, epidemic models, catalytic reactions, branching-annihilation RW, some growth processes.

Universality class: the contact process

Lattice model of interacting particles (belongs to DP class): $s_i(t) \in \{0, 1\}$, asynchronous update

$$\begin{cases} \Pr[s_i(t+1) = 1 | s_i(t) = 0] = 1 - \prod_{j \in \partial i} (1 - \lambda \mathbb{I}[s_j(t) = 1]) \\ \Pr[s_i(t+1) = 0 | s_i(t) = 1] = 1 \end{cases} \quad (9)$$

or with rates in continuous time (d dim)

$$\begin{cases} W[0 \rightarrow 1 | n] = \lambda n \\ W[1 \rightarrow 0 | n] = 1 \end{cases} \quad (10)$$

Monte Carlo simulations in $1 + 1$ dim: continuous transition with critical point at λ_c .

MF approximation for contact process

Dynamics of the average local density $\rho(\vec{x}, t)$ of active sites is given by

$$\frac{\partial}{\partial t} \rho(\vec{x}, t) = D \nabla^2 \rho(\vec{x}, t) - \rho(\vec{x}, t) + \lambda \rho(\vec{x}, t) (1 - \rho(\vec{x}, t)) \quad (11)$$

$$= D \nabla^2 \rho(\vec{x}, t) + (\lambda - 1) \rho(\vec{x}, t) - \lambda \rho(\vec{x}, t)^2 \quad (12)$$

The stationary solution is spatially homogenous

$$\rho^{\text{st}} = \begin{cases} 0 & \lambda < \lambda_c = 1 \\ \frac{\lambda-1}{\lambda} & \lambda > \lambda_c, \end{cases} \implies \beta^{MF} = 1 \quad (13)$$

The density decays in the inactive phase as

$$\rho(t) \sim e^{-|\lambda - \lambda_c|t} \sim e^{-t/\xi_{\parallel}}, \quad \nu_{\parallel}^{MF} = 1. \quad (14)$$

MF approximation for contact process

Close to criticality, spreading process should be invariant under length rescaling $\vec{x} = \ell \vec{x}'$ and rescaling of $\kappa = \lambda - \lambda_c$,

$$t = \ell^z t', \quad \kappa = \ell^{-1/\nu_\perp} \kappa', \quad \rho = \ell^{-\beta/\nu_\perp} \rho' \quad (15)$$

Rescaled equation

$$\begin{aligned} \ell^{-z} \frac{\partial}{\partial t'} \ell^{-\beta/\nu_\perp} \rho'(\vec{x}', t') &= D \ell^{-2} \nabla^2 \ell^{-\beta/\nu_\perp} \rho'(\vec{x}', t') \\ &\quad + \ell^{-1/\nu_\perp} \kappa' \ell^{-\beta/\nu_\perp} \rho'(\vec{x}', t') - \lambda \ell^{-2\beta/\nu_\perp} \rho'(\vec{x}', t')^2 \end{aligned} \quad (16)$$

is invariant if

$$z + \frac{\beta}{\nu_\perp} = \frac{1}{\nu_\perp} + \frac{\beta}{\nu_\perp} = 2 + \frac{\beta}{\nu_\perp} = \frac{2\beta}{\nu_\perp} \implies \nu_\perp^{MF} = \frac{1}{2}, \quad \beta^{MF} = 1, \quad z = 2. \quad (17)$$

Langevin equation for the DP class

$$\tau \frac{\partial}{\partial t} \rho(\vec{x}, t) = D \nabla^2 \rho(\vec{x}, t) + \kappa \rho(\vec{x}, t) - \lambda \rho(\vec{x}, t)^2 + \eta(\vec{x}, t) \quad (18)$$

with

$$\langle \eta(\vec{x}, t) \rangle = 0 \quad (19)$$

$$\langle \eta(\vec{x}, t) \eta(\vec{x}', t') \rangle = \gamma \rho(\vec{x}, t) \delta(\vec{x} - \vec{x}') \delta(t - t') \quad (20)$$

Noise amplitude $\sim \rho^{1/2}$ comes from central limit theorem. We added τ for generality.

REMARK: The noise appearing here is a *multiplicative noise*, ensuring that no fluctuations occur if $\rho(\vec{x}, t) = 0$.

REMARK: Noise term is not always correctly described by phenomenological assumptions (e.g. pair-annihilation process)

Field Theory for DP: the action

We employ the response-functional (MSR) method, we obtain the functional integral

$$Z \propto \int D\eta P[\eta] \int D\rho \det J[\rho] \delta \left(\tau \partial_t \rho - D \nabla^2 \rho - \kappa \rho + \lambda \rho^2 - \eta \right) \quad (21)$$

$$\propto \int D\eta P[\eta] \int D\rho D\tilde{\rho} e^{-\int d\vec{x} dt [\tilde{\rho} (\tau \partial_t \rho - D \nabla^2 \rho - \kappa \rho + \lambda \rho^2 - \eta)]} \quad (22)$$

$$\begin{aligned} &\propto \int D\rho D\tilde{\rho} e^{-\int d\vec{x} dt [\tilde{\rho} (\tau \partial_t \rho - D \nabla^2 \rho - \kappa \rho + \lambda \rho^2)]} \\ &\quad \times \int D\eta e^{\int d\vec{x} dt [\tilde{\rho} \eta - \frac{1}{2\gamma\rho} \eta^2]} \end{aligned} \quad (23)$$

$$\propto \int D\rho D\tilde{\rho} e^{-\int d\vec{x} dt [\tilde{\rho} (\tau \partial_t \rho - D \nabla^2 \rho - \kappa \rho + \lambda \rho^2)]} e^{\frac{\gamma}{2} \int d\vec{x} dt \rho \tilde{\rho}^2} \quad (24)$$

we used $\det J[\rho] = 1$ (Ito regularization), introduced a response field $\tilde{\rho}(\vec{x}, t)$ and averaged over Gaussian noise field $\eta(\vec{x}, t)$.

Field Theory for DP: the action

Setting $\rho = \sqrt{\frac{\gamma}{2\lambda}}\phi$ and $\tilde{\rho} = \sqrt{\frac{2\lambda}{\gamma}}\tilde{\phi}$,

$$\lambda\tilde{\rho}\rho^2 = \lambda\sqrt{\frac{2\lambda}{\gamma}}\frac{\gamma}{2\lambda}\tilde{\phi}\phi^2 = \sqrt{\frac{\lambda\gamma}{2}}\tilde{\phi}\phi^2 \quad (25)$$

$$\frac{\gamma}{2}\tilde{\rho}^2\rho = \frac{\gamma}{2}\frac{2\lambda}{\gamma}\sqrt{\frac{\gamma}{2\lambda}}\tilde{\phi}^2\phi = \sqrt{\frac{\lambda\gamma}{2}}\tilde{\phi}^2\phi \quad (26)$$

Then we define $\Gamma/2 = \sqrt{\lambda\gamma/2}$.

$$Z \propto \int D\phi D\tilde{\phi} e^{-\int d\vec{x}dt [\tilde{\phi}(\tau\partial_t - D\nabla^2 - \kappa)\phi + \frac{\Gamma}{2}(\tilde{\phi}\phi^2 - \phi\tilde{\phi}^2)]} \quad (27)$$

$$= \int D\phi D\tilde{\phi} e^{-S[\phi, \tilde{\phi}]} \quad (28)$$

REMARK: It is also known in particle physics as *Reggeon field theory*.

Field Theory for DP: the action

Dimensional analysis of the action

$$S[\phi, \tilde{\phi}] = \int d\vec{x} dt \left[\tilde{\phi} \left(\tau \partial_t - D \nabla^2 - \kappa \right) \phi + \frac{\Gamma}{2} \left(\tilde{\phi} \phi^2 - \phi \tilde{\phi}^2 \right) \right] \quad (29)$$

Since $[x] = \Lambda^{-1}$, $[t] = [x]^z = \Lambda^{-2}$ (also $[\tau] = 1$), then

$$[\tilde{\phi} \partial_t \phi] = [D \tilde{\phi} \nabla^2 \phi] = [\kappa \tilde{\phi} \phi] = [x]^{-d} [t]^{-1} = \Lambda^{d+2} \quad (30)$$

which means $[\tilde{\phi} \phi] = \Lambda^d$, $[D] = 1$ and $[\kappa] = \Lambda^2$. Then

$$[\tilde{\phi} \phi^2] = [\phi \tilde{\phi}^2] \implies [\tilde{\phi}] = [\phi] = \Lambda^{d/2}, \quad (31)$$

and $[\Gamma] = \Lambda^{d+2-3d/2} = \Lambda^{2-d/2}$.

The upper critical dimension of DP (Reggeon) field theory is $d_{uc} = 4$.

Field Theory for DP: the action

Rescaling $\vec{x} \rightarrow \vec{x}' = \vec{x}/\ell$, we have

$$t \rightarrow t' = \ell^{-z} t, \quad \phi \rightarrow \phi' = \ell^{-\chi} \phi, \quad \tilde{\phi} \rightarrow \tilde{\phi}' = \ell^{-\chi} \tilde{\phi}, \quad (32)$$

$$\begin{aligned} S = & \int d\vec{x}' dt' \left[\tilde{\phi}'(\vec{x}', t') \left(\tau \ell^{d+2\chi} \partial_{t'} + D \ell^{2\chi+d+z-2} \nabla_{\vec{x}'}^2 - \ell^{2\chi+d+z} \kappa \right) \phi'(\vec{x}', t') \right] \\ & + \frac{\Gamma \ell^{3\chi+d+z}}{2} \int d\vec{x} dt' \left[\tilde{\phi}'(\vec{x}', t') \phi'(\vec{x}', t') \phi'(\vec{x}', t') - \phi'(\vec{x}', t') \tilde{\phi}'(\vec{x}', t') \tilde{\phi}'(\vec{x}', t') \right] \end{aligned}$$

Rescaled parameters

$$\tau' = \ell^{d+2\chi} \tau \quad (33)$$

$$D' = \ell^{2\chi+d+z-2} D \quad (34)$$

$$\kappa' = \ell^{2\chi+d+z} \kappa \quad (35)$$

$$\Gamma' = \ell^{3\chi+d+z} \Gamma. \quad (36)$$

Field Theory for DP: the action

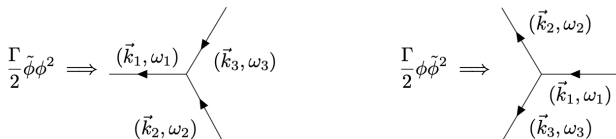
In the Fourier space

$$\begin{aligned} S = & \int \frac{d\vec{k}}{(2\pi)^d} \int \frac{d\omega}{2\pi} [\tilde{\phi}(\vec{k}, \omega) (-i\tau\omega + Dk^2 - \kappa) \phi(-\vec{k}, -\omega)] \\ & + \frac{\Gamma}{2} \int \frac{d\vec{k}_1 d\vec{k}_2}{(2\pi)^{2d}} \int \frac{d\omega_1 d\omega_2}{(2\pi)^2} [\tilde{\phi}(\vec{k}_1, \omega_1) \phi(\vec{k}_2, \omega_2) \phi(-\vec{k}_1 - \vec{k}_2, -\omega_1 - \omega_2) \\ & - \phi(\vec{k}_1, \omega_1) \tilde{\phi}(\vec{k}_2, \omega_2) \tilde{\phi}(-\vec{k}_1 - \vec{k}_2, -\omega_1 - \omega_2)] . \end{aligned} \quad (37)$$

The free propagator is

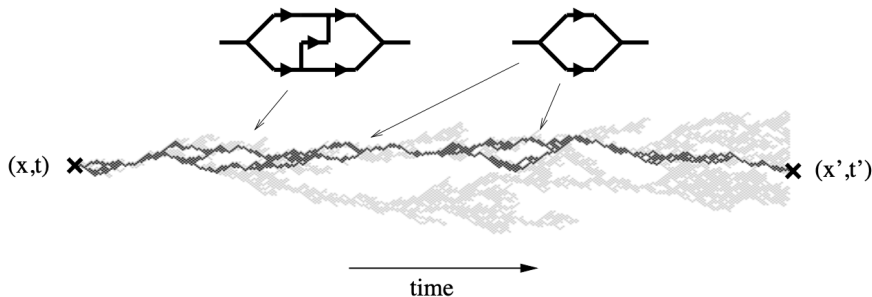
$$G_0(\vec{k}, \omega) = \frac{1}{-i\omega\tau + Dk^2 - \kappa} . \quad (38)$$

Vertices are



Field Theory for DP: physical interpretation of diagrams

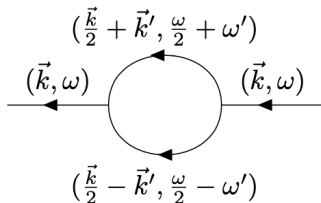
The pair correlation (or connectedness) function $C(\vec{x}, t, \vec{x}', t')$, i.e. probability that two sites are connected by a directed path of open bonds, can be represented as a sum over paths with branching and annihilation points.



Field Theory for DP: one-loop RG calculations

First-order correction to the propagator (renormalization of τ, D, κ):

$$G_{<}(\vec{k}, \omega) = G_{0,<}(\vec{k}, \omega) - 2 \left(\frac{\Gamma}{2} \right)^2 G_{0,<}(\vec{k}, \omega)^2 \quad (39)$$
$$\times \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \int \frac{d\omega'}{2\pi} G_{0,>}(\frac{\vec{k}}{2} + \vec{k}', \frac{\omega}{2} + \omega') G_{0,>}(\frac{\vec{k}}{2} - \vec{k}', \frac{\omega}{2} - \omega') + o(\Gamma^2)$$



Field Theory for DP: one-loop RG calculations

$$G_{<}(\vec{k}, \omega) = G_{0,<}(\vec{k}, \omega) \left[1 - 2 \left(\frac{\Gamma}{2} \right)^2 G_{0,<}(\vec{k}, \omega) \right. \\ \left. \times \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \int \frac{d\omega'}{2\pi} G_{0,>}(\frac{\vec{k}}{2} + \vec{k}', \frac{\omega}{2} + \omega') G_{0,>}(\frac{\vec{k}}{2} - \vec{k}', \frac{\omega}{2} - \omega') \right] \quad (40)$$

$$G_{<}(\vec{k}, \omega)^{-1} = G_{0,<}(\vec{k}, \omega)^{-1} \\ + \frac{\Gamma^2}{2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \int \frac{d\omega'}{2\pi} G_{0,>}(\frac{\vec{k}}{2} + \vec{k}', \frac{\omega}{2} + \omega') \\ \times G_{0,>}(\frac{\vec{k}}{2} - \vec{k}', \frac{\omega}{2} - \omega') \quad (41)$$

$$-i\omega\tau_1 + D_1 k - \kappa_1 = -i\omega\tau + Dk^2 - \kappa + I_P(\vec{k}, \omega) \quad (42)$$

$$= -i\omega\tau + Dk^2 - \kappa + I_P(0, 0) \quad (43)$$

$$+ i\omega \left. \frac{\partial I_P(\vec{k}, \omega)}{\partial i\omega} \right|_{\omega=0, k=0} + k^2 \left. \frac{\partial I_P(\vec{k}, \omega)}{\partial k^2} \right|_{\omega=0, k^2=0} + \dots$$

Field Theory for DP: one-loop RG calculations

$$\begin{aligned} I_P(\vec{k}, \omega) &= \frac{\Gamma^2}{2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \int \frac{d\omega'}{2\pi} G_{0,>}(\frac{\vec{k}}{2} + \vec{k}', \frac{\omega}{2} + \omega') G_{0,>}(\frac{\vec{k}}{2} - \vec{k}', \frac{\omega}{2} - \omega') \\ &= \frac{\Gamma^2}{2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \frac{d\omega'}{2\pi} \frac{1}{A^+ - i\tau\omega'} \frac{1}{A^- + i\tau\omega'} \end{aligned} \quad (44)$$

where $A^{\pm} = D \left(\frac{\vec{k}}{2} \pm \vec{k}' \right)^2 - \kappa - i\tau \frac{\omega}{2}$. Using residues (pole $\omega' = iA^{\pm}/\tau$),

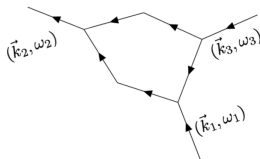
$$I_P(\vec{k}, \omega) = \frac{\Gamma^2}{2} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}'}{(2\pi)^{d+1}} \frac{2\pi i}{i\tau (A^+ + A^-)} \quad (45)$$

$$\approx \frac{\Gamma^2}{4\tau} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \frac{1}{\frac{Dk^2}{4} + Dk'^2 - \kappa - i\tau \frac{\omega}{2}} \quad (46)$$

$$\approx \frac{\Gamma^2}{4\tau} \frac{K_d \Lambda^d}{\frac{Dk^2}{4} + D\Lambda^2 - \kappa - i\tau \frac{\omega}{2}} \left(1 - \frac{1}{\ell} \right) \quad (47)$$

Field Theory for DP: one-loop RG calculations

First-order correction to interaction vertices (renormalization of Γ):
diagrams of the type



$$\frac{\Gamma_1}{2} = \frac{\Gamma}{2} - 8 \left(\frac{\Gamma}{2} \right)^3 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \frac{d\omega}{2\pi} G_{0,>}(\vec{k}, \omega)^2 G_{0,>}(-\vec{k}, -\omega) \quad (48)$$

Field Theory for DP: one-loop RG calculations

$$\Gamma_1 = \Gamma - 2\Gamma^3 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \frac{d\omega}{2\pi} \frac{1}{(Dk^2 - \kappa - i\tau\omega)^2} \frac{1}{(Dk^2 - \kappa + i\tau\omega)} \quad (49)$$

$$= \Gamma - 2\Gamma^3 \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \frac{2\pi i}{4i\tau (Dk^2 - \kappa)^2} \quad (50)$$

$$\approx \Gamma - \frac{\Gamma^3 K_d \Lambda^d}{2\tau (D\Lambda^2 - \kappa)^2} \left(1 - \frac{1}{\ell}\right) \quad (51)$$

the ω integration performed computing residues at pole $i\tau\omega = Dk^2 - \kappa$.

Field Theory for DP: one-loop RG calculations

Expanding $\ell = e^{\delta s} \approx 1 + \delta s$, the RG flow equations at one-loop read

$$\frac{\partial \tau}{\partial s} = \left[(2\chi + d) - \frac{\Gamma^2 K_d \Lambda^d}{8\tau (D\Lambda^2 - \kappa)^2} \right] \tau = [2\chi + d - 2S_1] \tau \quad (52)$$

$$\frac{\partial D}{\partial s} = \left[(2\chi + d + z - 2) - \frac{\Gamma^2 K_d \Lambda^d}{16\tau (D\Lambda^2 - \kappa)^2} \right] D = [2\chi + d + z - 2 - S_1] D \quad (53)$$

$$\frac{\partial \kappa}{\partial s} = \left[(2\chi + d + z) - \frac{\Gamma^2 K_d \Lambda^d}{4\tau \kappa (D\Lambda^2 - \kappa)} \right] \kappa = [2\chi + d + z - S_2] \kappa \quad (54)$$

$$\frac{\partial \Gamma}{\partial s} = \left[(3\chi + d + z) - \frac{\Gamma^2 K_d \Lambda^d}{2\tau (D\Lambda^2 - \kappa)^2} \right] \Gamma = [3\chi + d + z - 8S_1] \Gamma \quad (55)$$

where

$$S_1 = \frac{\Gamma^2 K_d \Lambda^d}{16\tau (D\Lambda^2 - \kappa)^2}, \quad S_2 = \frac{\Gamma^2 K_d \Lambda^d}{4\tau \kappa (D\Lambda^2 - \kappa)}. \quad (56)$$

Field Theory for DP: one-loop RG calculations

Setting τ and D to be invariant under RG, we find (at $d = 4 - \epsilon$)

$$2\chi + 4 - \epsilon = 2S_1, \quad 2\chi + d + z - 2 = S_1 \quad (57)$$

$$\frac{\partial \kappa}{\partial s} = \kappa (2 + S_1 - S_2) \quad (58)$$

$$\frac{\partial \Gamma}{\partial s} = \Gamma \left(\frac{\epsilon}{2} - 6S_1 \right) \quad (59)$$

with (non-trivial) fixed point $S_1^* = \frac{\epsilon}{12}$, $S_2^* = 2 + \frac{\epsilon}{12}$, corresponding to

$$\kappa^* = 4 \frac{S_1^*}{S_2^*} D \Lambda^2 \approx \frac{D \Lambda^2}{6} \epsilon \quad (60)$$

$$\Gamma^{2,*} = \frac{16\tau (D \Lambda^2 - \kappa)^2}{12K_d \Lambda^4} \epsilon \approx \frac{4\tau D^2}{3K_d} \epsilon \quad (61)$$

Field Theory for DP: critical exponents

Linearizing the RG equations around (κ^*, Γ^*) , we get a stability matrix with eigenvalues

$$y_\kappa = 2 - \frac{\epsilon}{4} + O(\epsilon^2), \quad y_\Gamma = -\epsilon + O(\epsilon^2). \quad (62)$$

Noticing that $\kappa \sim \lambda - \lambda_c$, then $y_\kappa = \frac{1}{\nu_\perp}$, i.e. $\nu_\perp = \frac{1}{2(1-\epsilon/8)} \approx \frac{1}{2} (1 + \frac{\epsilon}{8})$

Then

$$\chi = S_1 - 2 + \frac{\epsilon}{2} = -2 + \frac{7}{12}\epsilon + O(\epsilon^2) \quad (63)$$

$$z = 2 - S_1 = 2 - \frac{\epsilon}{12} + O(\epsilon^2) \quad (64)$$

$$\beta = -\chi\nu_\perp = \left(1 - \frac{7}{24}\epsilon\right) \left(1 + \frac{\epsilon}{8}\right) = 1 - \frac{\epsilon}{6} + O(\epsilon^2) \quad (65)$$

$$\nu_\parallel = 1 + \frac{\epsilon}{12} + O(\epsilon^2). \quad (66)$$

Langevin equation for extrinsic multiplicative noise

- *Intrinsic noise*: generated by stochasticity internal to the system
- *Extrinsic noise*: generated by environmental fluctuations of parameters

Consider the phenomenological equation of the form

$$\frac{\partial}{\partial t}\phi(\vec{x}, t) = D\nabla^2\phi(\vec{x}, t) + \kappa\phi(\vec{x}, t) - \lambda\phi(\vec{x}, t)^b, \quad b \geq 2 \quad (67)$$

It could represent relaxation due to a thermodynamic force $-\Gamma\delta V/\delta\phi$ in a (bistable) potential

$$V[\phi] = -\frac{\kappa}{2\Gamma}\phi^2 + \frac{\lambda}{(b+1)\Gamma}\phi^{b+1}. \quad (68)$$

Langevin equation for extrinsic multiplicative noise

Stochastic effects due to *environmental fluctuations* of parameters, e.g.

$$\kappa(\vec{x}, t) = \kappa_0 + \tilde{\sigma}\eta(\vec{x}, t), \quad (69)$$

with additive noise

$$\begin{aligned} \langle \eta(\vec{x}, t) \rangle &= 0 \\ \langle \eta(\vec{x}, t) \eta(\vec{x}', t') \rangle &= \delta(t - t') \delta(\vec{x} - \vec{x}') \end{aligned} \quad (70)$$

The field ϕ is described by a Langevin equation with extrinsic multiplicative noise (EMN),

$$\tau \frac{\partial}{\partial t} \phi(\vec{x}, t) = D \nabla^2 \phi(\vec{x}, t) + \kappa_0 \phi(\vec{x}, t) - \lambda \phi(\vec{x}, t)^b + \tilde{\sigma} \phi(\vec{x}, t) \eta(\vec{x}, t). \quad (71)$$

REMARK: Extrinsic noise amplitude $\sim \phi$ not $\sim \phi^{1/2}$ as from CLT for intrinsic noise.

Field theory for the extrinsic multiplicative noise

Employing (MSR) response functional formalism, we obtain the action

$$S[\phi, \tilde{\phi}] = \int d\vec{x} dt \left[\tilde{\phi} \left(\tau \partial_t - D \nabla^2 - \kappa_0 \right) \phi + \lambda \tilde{\phi} \phi^b - \sigma^2 \tilde{\phi}^2 \phi^2 \right] \quad (72)$$

with $\sigma^2 = \tilde{\sigma}^2/2$

From dimensional analysis,

$$[\tilde{\phi} \phi] = \Lambda^d, \quad [D] = [\tau] = 1, \quad [\kappa_0] = \Lambda^2, \quad (73)$$

which implies

$$[\sigma^2] = \Lambda^{-2d+d+2} = \Lambda^{2-d} \quad (74)$$

and the upper critical dimension is $d_{uc} = 2$ (indep. of b).

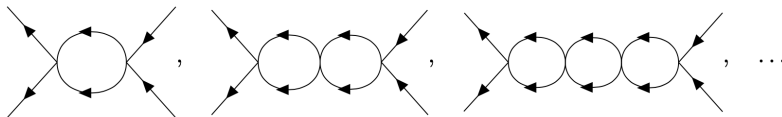
Field theory for the extrinsic multiplicative noise

There are two non-quadratic vertices,

$$-\sigma^2 \tilde{\phi}^2 \phi^2 \implies \text{crossing lines} \qquad \lambda \tilde{\phi} \phi^b \implies \left. \begin{array}{c} \text{trivalent vertex} \\ \text{with } b \text{ external lines} \end{array} \right\} b=2$$

REMARK: Corrections to the propagator are not possible (so τ, D, κ_0 are not renormalized).

REMARK: Both vertices are renormalized due to the noise vertex, with corrections of type



Renormalization of the noise coupling

Let us compute the 1-loop integral (over fast fields)

$$I = \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \int \frac{d\omega}{2\pi} G_{0,>}(\vec{k}, \omega) G_{0,>}(-\vec{k}, -\omega) \quad (75)$$

$$= \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \int \frac{d\omega}{2\pi} \frac{1}{i\tau\omega + Dk^2 - \kappa_0} \frac{1}{-i\tau\omega + Dk^2 - \kappa_0} \quad (76)$$

$$= \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \int \frac{d\omega}{2\pi} \frac{1}{A + i\tau\omega} \frac{1}{A - i\tau\omega} \quad (77)$$

$$= \frac{1}{2\tau} \int_{\Lambda/\ell}^{\Lambda} \frac{d\vec{k}}{(2\pi)^d} \frac{1}{Dk^2 - \kappa_0} \quad (78)$$

$$\approx \frac{K_d}{2\tau} \frac{\Lambda^d}{D\Lambda^2 - \kappa_0} \left(1 - \frac{1}{\ell}\right) \quad (79)$$

$$\approx \frac{K_d \Lambda^{-\epsilon}}{2\tau D} \left(1 - \frac{1}{\ell}\right), \quad \epsilon = 2 - d \quad (80)$$

RG flow equations

First-order correction to σ^2 is

$$\sigma'^2 = \sigma^2 + \left(-\sigma^2\right)^2 l \quad (81)$$

Close to $d_{\text{uc}} = 2$, it is convenient to introduce the non-dimensional parameter $u = \sigma^2 \Lambda^{-\epsilon}$.

The renormalized parameter is given by

$$u(\ell) = \ell^{2-d} u' \quad (82)$$

$$= \ell^{2-d} u [1 + u \Lambda^\epsilon l] \quad (83)$$

$$= \ell^\epsilon u \left[1 + u \frac{K_{2-\epsilon}}{2\tau D} \left(1 - \frac{1}{\ell} \right) \right] \quad (84)$$

$$\approx (1 + \epsilon \delta s) u (1 + u \alpha \delta s) \quad (85)$$

where we expanded $\ell = e^s$ for small s and set $\alpha = K_2/(2\tau D)$.

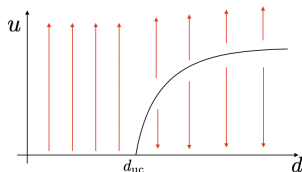
Noise-driven phase transition

The RG flow equation reads

$$\frac{du}{ds} = \epsilon u + \alpha u^2.$$

Fixed points:

- for $d < 2$, only $u = 0$ (weak-noise limit) and $u = \infty$ (strong-noise limit) exist. The weak-noise fixed point is unstable.
- for $d > 2$, it exists a critical value $u_* = -\epsilon/\alpha$, which separates (“phase transition”) basins of attraction of two stable fixed points describing weak-noise and strong-noise regimes.



A “super-renormalizable” theory

Let us consider the integral over whole momentum spectrum,

$$I_0 = \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \int \frac{d\omega}{2\pi} G_0(\vec{k}, \omega) G_0(-\vec{k}, -\omega) \quad (86)$$

We can formally resum the whole series of diagrams,

$$\sigma_R^2 = \sigma^2 + (-\sigma^2)^2 I_0 - (-\sigma^2)^3 I_0^2 + \dots \quad (87)$$

$$= \sigma^2 \left[1 + \sigma^2 I_0 + (\sigma^2 I_0)^2 + \dots \right] \quad (88)$$

$$= \frac{\sigma^2}{1 - \sigma^2 I_0}, \quad (89)$$

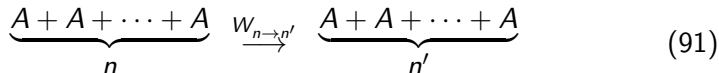
I_0 is finite for $d > 2$ (“super-renormalizable” field theory) and IR divergent for $d \leq 2$.

REMARK: The other vertex can be also renormalized accordingly,

$$\lambda_R = \lambda \left(1 + \sigma^2 I_0 + \sigma^4 I_0^2 + \dots \right) = \frac{\lambda}{1 - \sigma^2 I_0} = \frac{\lambda}{\sigma^2} \sigma_R^2. \quad (90)$$

Reaction-diffusion: from master equation to field theory

Consider a “well-mixed” (0-dim) interacting particle system with n particles which undergo reaction processes, represented by stochastic transitions of the type



The probability $P(n, t)$ that the system contains n particles at time t is given by the solution of the *master equation*,

$$\partial_t P(n, t) = \sum_{n'} [W_{n' \rightarrow n} P(n', t) - W_{n \rightarrow n'} P(n, t)], \quad (92)$$

with some initial condition $P(n, 0) = P_0(n)$, e.g. $P_0(n) = \delta_{n, n_0}$.

Master equation for reaction processes

Consider a process containing reactions



corresponding to the transition rates

$$W_{n \rightarrow n'} = \mu n \delta_{n', n-1} + \lambda n \delta_{n', n+1} + \frac{\sigma}{2} n(n-1) \delta_{n', n-1}. \quad (94)$$

The master equation reads

$$\begin{aligned} \partial_t P(n, t) = & \mu(n+1)P(n+1, t) - \mu n P(n, t) \\ & + \lambda(n-1)P(n-1, t) - \lambda n P(n, t) \\ & + \frac{\sigma}{2}(n+1)n P(n+1, t) - \frac{\sigma}{2}n(n-1)P(n, t) \end{aligned} \quad (95)$$

Master equation for reaction processes

It is convenient to introduce a “step” operator $\hat{E}_m$ on the discrete n -space, such that

$$\hat{E}_m f(n) = f(n + m). \quad (96)$$

Notice that, using Taylor expansion,

$$f(n + m) = f(n) + mf'(n) + \frac{m^2}{2}f''(n) + \dots \quad (97)$$

$$= e^{m\partial_n} f(n) \quad (98)$$

The master equation for previous example becomes

$$\partial_t P(n, t) = \left[\mu (E_{+1} - 1) n + \frac{\sigma}{2} (E_{+1} - 1) n(n-1) + \lambda (E_{-1} - 1) n \right] P(n, t) \quad (99)$$

$$= \left[\mu (e^{\partial_n} - 1) n + \frac{\sigma}{2} (e^{\partial_n} - 1) n(n-1) + \lambda (e^{-\partial_n} - 1) n \right] P(n, t) \quad (100)$$

$$= -\tilde{H}_R [\partial_n, n] P(n, t). \quad (101)$$

Master equation for reaction processes

Let us define a “momentum” operator p , such that its n -space representation is $p = -i\hbar\partial_n$, it satisfies canonical commutation relations

$$[n, p] = np - pn \quad (102)$$

$$= -ni\hbar\frac{\partial}{\partial n} + i\hbar\frac{\partial}{\partial n}n \quad (103)$$

$$= -ni\hbar\frac{\partial}{\partial n} + i\hbar\left(1 + n\frac{\partial}{\partial n}\right) \quad (104)$$

$$= i\hbar \quad (105)$$

The “reaction Hamiltonian” $\tilde{H}_R$ becomes

$$\tilde{H}_R[p, q] = -\mu \left(e^{\frac{i}{\hbar}p} - 1\right) q - \frac{\sigma}{2} \left(e^{\frac{i}{\hbar}p} - 1\right) q(q-1) - \lambda \left(e^{-\frac{i}{\hbar}p} - 1\right) q \quad (106)$$

Path-integral formulation of quantum mechanics

Consider Schrodinger equation

$$i\hbar \frac{d}{dt} |\Phi\rangle = H |\Phi\rangle \quad (107)$$

with formal solution

$$|\Phi(t)\rangle = e^{-iHt/\hbar} |\Phi(0)\rangle = U(t) |\Phi(0)\rangle \quad (108)$$

Using *Feynman-Kac formula* for the path integral representation of quantum mechanics, the probability amplitude of a quantum particle is

$$\langle x', t' | x, t \rangle = \langle x' | U(t' - t) | x \rangle \quad (109)$$

$$= \int_{q(t)=x}^{q(t')=x'} Dq Dp e^{\frac{i}{\hbar} \int_t^{t'} ds (p\dot{q} - H[p, q])} \quad (110)$$

where the functional integral is defined as the limit of multidimensional integrals and the measure is

$$Dq Dp = \lim_{N \rightarrow \infty} \prod_{k=0}^{N-1} \frac{dq_k dp_k}{2\pi\hbar}. \quad (111)$$

Path-integral formulation of quantum mechanics

Perform a “Wick rotation” to imaginary time $t \rightarrow \tau = it/\hbar$, we find

$$|\Phi(\tau)\rangle = e^{-H\tau} |\Phi(0)\rangle \quad (112)$$

and the amplitude

$$\langle x', t' | x, t \rangle = \langle x' | U(t' - t) | x \rangle \quad (113)$$

$$= \langle x' | e^{-H\tau} | x \rangle \quad (114)$$

$$= \int_{q(\tau)=x}^{q(\tau')=x'} Dq Dp e^{\int_{\tau}^{\tau'} ds (p \frac{i}{\hbar} \partial_{\tau} q - H[p, q])} \quad (115)$$

Setting $p \rightarrow -\frac{i}{\hbar} p$ (so that $\partial_n = -p$) we find

$$\langle x' | e^{-H\tau} | x \rangle = \int_{q(\tau)=x}^{q(\tau')=x'} Dq Dp e^{\int_{\tau}^{\tau'} ds (-p \partial_{\tau} q - H[p, q])} \quad (116)$$

Hamiltonian formalism for master equation

It follows that the formal solution of the master equation can be interpreted as

$$P(n, t) = \left\langle n | e^{-t\tilde{H}_R} | n_0 \right\rangle \quad (117)$$

$$= \int_{q(0)=n_0}^{q(t)=n} Dq Dp e^{-\int_0^t ds (p\dot{q} - H_R[p, q])} \quad (118)$$

with reaction Hamiltonian

$$H_R[p, q] = \mu (e^{-p} - 1) q + \frac{\sigma}{2} (e^{-p} - 1) q(q - 1) + \lambda (e^p - 1) q. \quad (119)$$

REMARK: The commutation relation is given by

$$[q, p] = qp - pq \quad (120)$$

$$= -q \frac{\partial}{\partial q} + \frac{\partial}{\partial q} q \quad (121)$$

$$= 1 \quad (122)$$

Hamiltonian formalism for master equation

Fokker-Planck approx.: expand $e^{\pm p}$ around $p = 0$, and truncate at $O(p^2)$,

$$H_R[p, q] = \mu \left(-p + \frac{1}{2} p^2 \right) q + \frac{\sigma}{2} \left(-p + \frac{1}{2} p^2 \right) q(q-1) + \lambda \left(p + \frac{1}{2} p^2 \right) q \quad (123)$$

$$= -\mu p q + \frac{\mu}{2} p^2 q - \frac{\sigma}{2} p q(q-1) + \frac{\sigma}{4} p^2 q(q-1) + \lambda p q + \frac{\lambda}{2} p^2 q \quad (124)$$

$$= -(\mu - \sigma/2 - \lambda) p q + \frac{(\mu + \lambda - \sigma/2)}{2} p^2 q - \frac{\sigma}{2} p q^2 + \frac{\sigma}{4} p^2 q^2 \quad (125)$$

The bulk behavior of the system is described by the action

$$S[p, q] = \int dt (p \dot{q} - H_R[p, q]) \quad (126)$$

$$= \int dt \left[p \dot{q} + (\mu - \sigma/2 - \lambda) p q - \frac{(\mu + \lambda - \sigma/2)}{2} p^2 q + \frac{\sigma}{2} p q^2 - \frac{\sigma}{4} p^2 q^2 \right] \quad (127)$$

$$= \int dt \left[p (\partial_t + (\mu - \sigma/2 - \lambda)) q - \frac{(\mu + \lambda - \sigma/2)}{2} p^2 q + \frac{\sigma}{2} p q^2 - \frac{\sigma}{4} p^2 q^2 \right] \quad (128)$$

Hamiltonian formalism for master equation

Defining $\kappa = \mu - \sigma/2 - \lambda$ and for small κ (close to criticality)

$$S[p, q] = \int dt \left[p (\partial_t + \kappa) q - \frac{\kappa + 2\lambda}{2} p^2 q + \frac{\sigma}{2} p q^2 - \frac{\sigma}{4} p^2 q^2 \right] \quad (129)$$

$$\approx \int dt \left[p (\partial_t + \kappa) q - \lambda p^2 q + \frac{\sigma}{2} p q^2 - \frac{\sigma}{4} p^2 q^2 \right] \quad (130)$$

rescaling $p = \sqrt{\sigma/2\lambda}\tilde{\phi}$ and $q = \sqrt{2\lambda/\sigma}\phi$ and defining $\Gamma/2 = \sqrt{\frac{\sigma\lambda}{2}}$

we find

$$S[\tilde{\phi}, \phi] = \int dt \left[\tilde{\phi} (\partial_t + \kappa) \phi - \frac{\Gamma}{2} \tilde{\phi}^2 \phi + \frac{\Gamma}{2} \tilde{\phi} \phi^2 - \frac{\sigma}{4} \tilde{\phi}^2 \phi^2 \right] \quad (131)$$

Spatially-extended systems: diffusion

In discrete spatially-extended systems, described by particle-number vector $[n] = (n_1, \dots, n_N)$, we consider random-walk transitions,

$$(\dots, n_i, \dots, n_j, \dots) \xrightleftharpoons{\nu} (\dots, n_i - 1, \dots, n_j + 1, \dots)$$

for (i, j) neighboring sites.

The distribution $P((n_1, \dots, n_N), t)$ evolves under the master equation

$$\begin{aligned} \partial_t P([n], t) &= \nu \sum_{(i,j)} [(n_j + 1) P((\dots, n_i - 1, \dots, n_j + 1, \dots), t) \\ &\quad - n_i P((\dots, n_i, \dots, n_j, \dots), t)] \end{aligned} \quad (132)$$

$$= \nu \sum_{(i,j)} [(E_{j,+1} E_{i,-1} - 1) n_j + (E_{i,+1} E_{j,-1} - 1) n_i] P([n], t) \quad (133)$$

$$= \nu \sum_{(i,j)} [(e^{-p_j + p_i} - 1) n_j + (e^{-p_i + p_j} - 1) n_i] P([n], t) \quad (134)$$

Spatially-extended systems: diffusion

The corresponding “diffusion Hamiltonian” is

$$H_D [\{p_i, q_i\}] = \nu \sum_{(i,j)} [(e^{-p_j+p_i} - 1) q_j + (e^{-p_i+p_j} - 1) q_i], \quad (135)$$

$$= \frac{\nu}{2} \sum_i \sum_{j \in \partial i} [e^{-p_j} e^{+p_i} q_j - q_i] \quad (136)$$

For instance, in $d = 1$, we have

$$H_D [\{p_i, q_i\}] = \frac{\nu}{2} \sum_i [e^{-p_{i+1}} e^{+p_i} q_{i+1} - 2q_i + e^{-p_{i-1}} e^{+p_i} q_{i-1}] \quad (137)$$

$$= \frac{\nu}{2} \sum_i [e^{-p_i - \partial_x p_i \delta x} e^{+p_i} (q_i + \partial_x q_i \delta x) - 2q_i + e^{-p_i + \partial_x p_i \delta x} e^{+p_i} (q_i - \partial_x q_i \delta x)]$$

$$= \frac{\nu}{2} \sum_i [(1 - \partial_x p_i \delta x) (q_i + \partial_x q_i \delta x) - 2q_i + (1 + \partial_x p_i \delta x) (q_i - \partial_x q_i \delta x)] \quad (138)$$

$$= \frac{\nu \delta x^2}{2} \sum_i [-2\partial_x p_i \partial_x q_i] \quad (139)$$

In general dimension d : $H_D [\{p_i, q_i\}] = -D \sum_i [\nabla p_i \cdot \nabla q_i]$.

Field-theoretic action for DP

The *reaction-diffusion process* is described by the action,

$$S[\{p_i, q_i\}] = \int dt \sum_i \left[p_i \dot{q}_i + D \nabla p_i \cdot \nabla q_i - \tilde{H}_R[p_i, q_i] \right] \quad (140)$$

Taking the continuum limit, passing to fields $\phi(\vec{x}, t)$ and $\tilde{\phi}(\vec{x}, t)$, and integrating by parts, we obtain the field-theoretic action,

$$S[\phi, \tilde{\phi}] = \int d\vec{x} \int dt \left[\tilde{\phi} \left(\partial_t - D \nabla^2 + \kappa \right) \phi + \frac{\Gamma}{2} \left(\tilde{\phi} \phi^2 - \tilde{\phi}^2 \phi \right) - \Sigma \tilde{\phi}^2 \phi^2 \right] \quad (141)$$

REMARK: The fourth-order term is irrelevant at $d = d_{uc}$ indeed

$$[\Sigma] = \Lambda^{d+2-4d/2} = \Lambda^{2-d} \quad (142)$$

irrelevant for $d > 2$. Hence, we obtained exactly the action for DP.

- Directed Percolation (and absorbing phase-transitions):
H. Hinrichsen, *Nonequilibrium Critical Phenomena and Phase Transitions into Absorbing States*, Adv. in Phys. (2000), see <https://arxiv.org/abs/cond-mat/0001070>.
- Extrinsic Multiplicative Noise:
M.A. Munoz, *Multiplicative Noise in non-equilibrium phase transitions: A tutorial*, see <https://arxiv.org/pdf/cond-mat/0303650v2>.
- Derivations of Field Theory from master equation:
A. Kamenev, *Field Theory of Non-Equilibrium Systems*, Cambridge (2011), see Ch. 4 and Ch.8,
A. Altland and B. Simons, *Condensed Matter Field Theory*, Cambridge UP (2010), see Ch. 10.,
J. Cardy, *Field Theory and Non equilibrium Statistical Mechanics*, lecture notes, see J. Cardy's webpage at Oxford.